Hosam Semry

Software Engineer

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Profile

Backend-focused Software Engineer with experience building and optimizing production web systems using Python, Django, Django REST Framework, React, SQL Server, PostgreSQL, Redis, Celery, and Docker. Focused on API design, database performance, and scalable distributed processing.

Experience

Software Engineer — Global Technical Services *Jan 2026 – Present*

- Cut dashboard query latency from ∼3 minutes to under 250ms (∼700x) on a 5M+ record SQL Server database through indexing, query tuning, and Django ORM refactoring.
- Independently re-architected a legacy integration service into a multi-tenant Django/Celery platform that automates fuel-order ingestion, SCADA delivery reconciliation, and capacity monitoring for 14 oil distribution companies, handling ∼4,000 orders per day and cutting each processing cycle from 10 minutes to 20 seconds.
- Built resilient integrations with 14 external systems using Redis distributed locks, per-tenant circuit breakers, exponential backoff with jitter, and Fernet-encrypted credentials, with log-based monitoring.
- Designed a pluggable provider layer (database, REST, mock) using Adapter/Factory patterns, so new integrations can be added without touching core logic, backed by 56 unit and end-to-end tests.
- Implemented server-side pagination and multi-parameter filtering for data-heavy views and redesigned four dashboard modules around operational KPIs, eliminating browser freezes on large datasets.

Technologies: Python, Java Django REST Framework, Celery, Redis, SQL Server, JWT

Full-Stack Developer — Al-Alanda *May 2025 – Dec 2025*

- Developed and launched a multilingual (Arabic/English) showcase website with Django, integrating core business workflows.
- Improved backend performance via in-memory caching for a large product catalog, reducing redundant database queries.

Technologies: Python, Django

Projects

Club Management System GitHub

- Full-stack club management platform covering sales, tickets, events, inventory, and reporting with role-based authorization and async processing.

Technologies: Django REST Framework, PostgreSQL, Redis, Celery, React, Docker

Webify — Multi-Vendor E-Commerce Platform GitHub | Live Website

- Built a multi-vendor e-commerce platform with dedicated admin/seller dashboards, REST APIs, JWT and Google authentication, and Paymob payment integration.

Technologies: Django REST Framework, PostgreSQL, JWT, React, Redux, Bootstrap, Git

Education

B.Sc. Computer Engineering — Mansoura University *Sep 2019 – Jun 2024*

Python Full Stack Web Development — Information Technology Institute (ITI) *Nov 2024 – Apr 2025*

Skills

Languages	Python, JavaScript, TypeScript, Java, SQL
Backend	Django, Django REST Framework, Flask
Frontend	React, Next.js, Redux, Tailwind
Databases	SQL Server, PostgreSQL, MySQL, MongoDB
Infra & Tools	Docker, Git, Linux, AWS S3, Redis, Celery
Concepts	REST APIs, Caching, Design Patterns, SOLID Principles, Design Patterns